

Instructions

In this research-based assessment, you will explore a real-world problem using the concepts learned from your textbook. Your goal is to develop a practical solution that works within the given constraints. You don't need to implement the idea, but your proposal should be grounded in theory and backed by reliable sources. Summarize your findings in a clear and concise report, no longer than two pages (excluding References). Use Times New Roman font, size 11, single column, and single spacing.

Plagiarism Policy

We encourage original and creative thinking in your work. You may refer to textbooks, articles, or technical documentation; just be sure to cite all sources properly.

The accepted plagiarism (including AI plagiarism) threshold is $\leq 10\%$. Exceeding this may result in penalties.

Report Structure

Your report should follow this structured format:

- Introduction (3 - 4 sentences)
 - Briefly introduce the real-world system or scenario you are analyzing
 - Explain why this system is relevant to computer architecture
 - Mention the specific focus of your analysis
- Background / Related Work (1-2 paragraphs)
 - Summarize relevant textbook concepts
 - Mention the actual hardware architectures used in this scenario
 - Optionally, refer to industry practices or academic references
- Methodology / System Design
 - Propose an architectural strategy, optimization, or redesign based on your analysis
 - Include a diagram or table to explain the system architecture or data flow
 - Clearly state any assumptions
 - Justify how your solution improves performance, energy use, or system cost
 - Add some numeric comparison whenever possible
- Discussion (2 paragraphs)
 - Evaluate the impact of your proposal
 - Identify trade-offs or limitations
 - Mention any novel ideas or reasoning you explored during your solution design
 - *Evaluate the sustainability and impact of modern, fast-changing computer architecture and technology on society and the environment. (mandatory point - 1 single paragraph)*
(*discussion about power consumption, long-term sustainability, effect on the environment, etc.*)
- Conclusion (3-4 sentences)
 - Summarize the architectural insight gained from this problem

- Highlight how your approach benefits system design, sustainability, or user experience
 - If relevant, suggest future work
- References:
 - List any articles, technical specs, whitepapers, or datasheets used.
 - It is mandatory

Topic: Improving Parallel Scalability in Web Server Architectures

Small web platforms (generally used by local governments, schools, NGOs) increasingly rely on open-source web servers like Apache or Nginx to deliver services at scale. However, these systems often suffer from concurrency bottlenecks as user traffic grows.

Your task is to investigate how parallelism can improve performance in such web systems, focusing on software architecture, thread models, and memory contention.

In your report:

- Choose a real open-source server (e.g., Apache, Nginx, Node.js, or Express)
- Describe how it handles incoming requests:
- Process/thread-per-request vs thread pools vs event loops
- Shared memory regions (e.g., cache, session data)
- Identify a scalability bottleneck (e.g., locking, queue contention, memory pool thrashing)
- Propose a parallelism-aware fix:
- Switch to a lock-free queue
- Add per-thread memory queue
- Use I/O batching, epoll/kqueue for async workloads

Estimate improvement with metrics:

- E.g., reduce average response latency from 250 ms to <100 ms under 1000 concurrent users
- Reflect on how small architectural changes enable scalable services on low-cost infrastructure

Expected Outcome:

- Understand real-world concurrency models in web systems
- Diagnose performance limits tied to parallel thread behaviour or shared memory
- Propose targeted, architecture-informed enhancements
- Relate solutions to parallel processing principles: scalability, contention, latency
- Consider deployment feasibility in low-resource environments

<https://www.solo.io/topics/nginx>

<https://leapcell.io/blog/understanding-jemalloc>

<https://dev.to/codesensei/optimizing-nodejs-performance-best-practices-for-high-traffic-apps-4do9>

